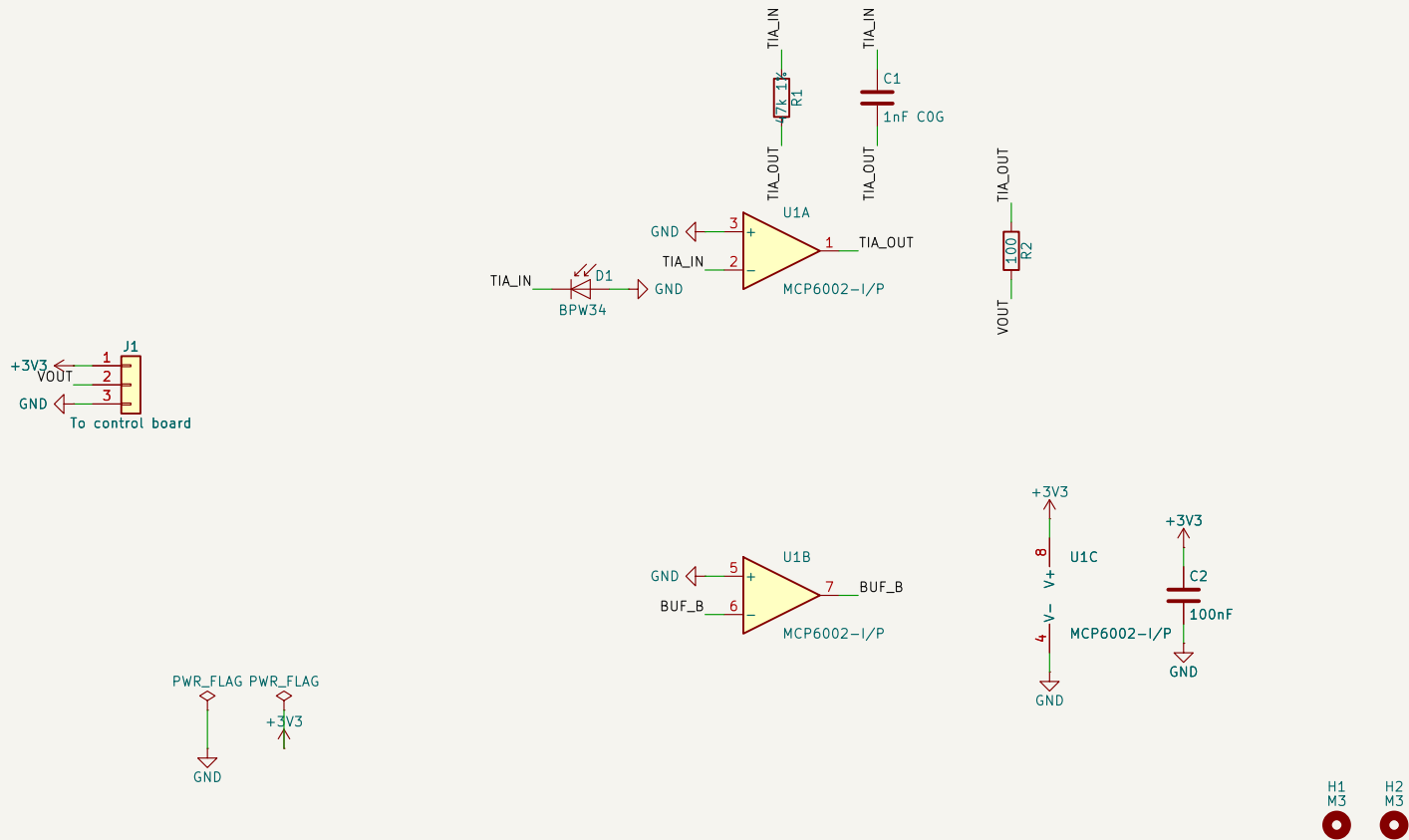


BPW34 at zero bias into an MCP6002 transimpedance amp. $V_{out} = I_{photo} \times R_f$.
47k || 1nF: full scale 70 uA (about 175 uW at 635 nm), 3.4 kHz bandwidth.
Build two: reference (control board J8 -> ADS1115 A0) and output (J9 -> A1).
If a channel saturates, fit a smaller R_f and change PD_*TIA_OHMS in config.h.



Half B is unused: wired as a grounded follower. R2 isolates the cable capacitance.